

Exercise 1.

a) If f is a polynomial algorithm for $A \in P$, and g is a polynomial algorithm for $B \in P$, then the following h is a polynomial algorithm for $A \cap B$:

$h(w) = \text{if } f(w) = 1 \text{ then (if } g(w) = 1 \text{ then } 1 \text{ else } 0) \text{ else } 0$



- $h(w) = 1 \iff w \in A \cap B$

- if $f = O(n^k)$ and $g = O(n^p)$, then $h = O(n^k + n^p) = O(n^{\max(k,p)})$, so h is polynomial

b) If f is a polynomial algorithm for $A \in P$, and g is a polynomial algorithm for $B \in P$, then the following h is a polynomial algorithm for $A \setminus B$:

$h(w) = \text{if } f(w) = 1 \text{ then (if } g(w) = 0 \text{ then } 1 \text{ else } 0) \text{ else } 0$

- $h(w) = 1 \iff w \in A \setminus B$

- if $f = O(n^k)$ and $g = O(n^p)$, then $h = O(n^k + n^p) = O(n^{\max(k,p)})$, so h is polynomial

c) Since $A \in NP$, there is a verifier $f_A(u, y_A)$ such that:

- $f_A(u, y_A) = 1$ iff $u \in A$ and y_A is a certificate for it.

Since $B \in NP$, there is a verifier $f_B(v, y_B)$ such that:

- $f_B(v, y_B) = 1$ iff $v \in B$ and y_B is a certificate for it.

The verifier $g(w, y)$ has to check whether $w \in A \cdot B$. Suppose $|w| = n$, then we split w at position k (where $0 \leq k \leq n$):

$w_1 = x_1 x_2 \dots x_k$: ~~to~~ be checked against A
 $w_2 = x_{k+1} x_{k+2} \dots x_n$ to be checked against B

The certificate y is of the form

$$y = k \# y_A \# y_B$$

$$g(w, y) = 1 \iff f_A(w_1, y_A) = 1 \text{ and } f_B(w_2, y_B) = 1$$

- g runs in polynomial time since both f_A , f_B runs in polynomial time.

- $g(w, y) = 1 \iff w \in A \cdot B$.

Exercise 2.

a) Since $B \in NP$, there is a polynomial verifier $V_B(w, y)$ such that:

$$V_B(w, y) = 1 \iff w \in B$$

and $f(a)$ can be computed in polynomial time $|a|$

$$L = \{a \in \{0, 1\}^* \mid f(a) \in B\}$$

The verifier $V_L(a, y)$ computes $f(a)$ and runs $V_B(f(a), y)$

$$V_L(a, y) = 1 \iff V_B(f(a), y) = 1$$

• If $a \in L$, then $f(a) \in B$. So there exists a certificate y such that $V_B(f(a), y) = 1$, and $V_L(a, y)$ will also return 1

• If $a \notin L$, then $f(a) \notin B$, so for all y $V_B(f(a), y) = 0$, and thus $V_L(a, y) = 0$

The size of the certificate y is polynomial in the size of $f(a)$

Verifier V_L also runs in polynomial time since $f(a)$ and V_B run in polynomial time.

b) If $A \in \text{NPH}$, then for any B in NP we have $B \leq_p A$. If $A \in \text{P}$, then we can polynomially decide B . That implies $\text{NP} \subseteq \text{P}$. Since we know that $\text{P} \subseteq \text{NP}$, this gives $\text{P} = \text{NP}$.

Exercise 4.

a)

$$\varphi_1 = (p \rightarrow \neg q) \wedge (p \wedge (q \vee \neg r)) \wedge (\neg p \rightarrow q) \wedge (\neg(q \wedge p \wedge r))$$

$$p \rightarrow \neg q \equiv \neg p \vee \neg q$$

$$\neg p \rightarrow q \equiv p \vee q$$

$$\neg(q \wedge p) \vee r \equiv (\neg q \vee \neg p \vee r)$$

$$\varphi_1 = (\neg p \vee \neg q) \wedge (p \wedge (q \vee \neg r)) \wedge (p \vee q) \wedge (\neg q \vee \neg p \vee r)$$

$$\text{CNF} = (\neg p \vee \neg q) \wedge p \wedge (q \vee \neg r) \wedge (p \vee q) \wedge (\neg q \vee \neg p \vee r)$$

This is satisfiable, for example $p=1, q=1, r=1$ satisfies all clauses.

$$b) \varphi_2 := (p \wedge \neg q) \vee (q \wedge r) \equiv (p \vee (q \wedge r)) \wedge (\neg q \vee (q \wedge r))$$

$$\equiv (p \vee q) \wedge (p \vee r) \wedge (\neg q \vee q) \wedge (\neg q \vee r)$$

It is satisfiable, for example $p=1, q=0, r=1$ satisfies it.

$$c) \varphi_2 := ((p \rightarrow q) \wedge (q \rightarrow r)) \rightarrow (p \rightarrow r)$$

$$p \rightarrow q \equiv \neg p \vee q$$

$$q \rightarrow r \equiv \neg q \vee r$$

$$p \rightarrow r \equiv \neg p \vee r$$

$$\varphi_2 \equiv ((\neg p \vee q) \wedge (\neg q \vee r)) \rightarrow (\neg p \vee r) \equiv$$

$$\neg((\neg p \vee q) \wedge (\neg q \vee r)) \vee (\neg p \vee r) \equiv$$

$$((p \wedge \neg q) \vee (q \wedge \neg r)) \vee (\neg p \vee r) \equiv$$

$$(p \vee \neg p \vee r) \wedge (\neg q \vee \neg p \vee r) \wedge (q \vee \neg p \vee r) \wedge (\neg r \vee \neg p \vee r)$$

It is satisfiable, for example $p=1, q=1, r=1$

Exercise 5.

a) This problem is in P. With only $\vee$ and the constant $\perp$, all formulas are either $\perp$, which is unsatisfiable, or a disjunction of variables, which is satisfiable as long as at least one variable is assigned the value 1. This can be checked in linear time.

b) This problem is in NPH. The set $\rightarrow, \neg$ is functionally complete, meaning all boolean functions can be expressed using these. The SAT problem remains as hard as the general SAT problem.

Exercise 6.

We know that f decides SAT in polynomial time. We can define g as follows

- if $f(\varphi) = 0$, then stop and return 0, otherwise continue with 1.
- compute the atoms in $\varphi := x_1, \dots, x_m$
- put $\varphi' := \varphi[x_m := \neg x_m]$
- if $f(\varphi') = 1$ then (put $a(x_m) := 1$) else (put $a(x_m) := 0$ and put $\varphi' := \varphi[x_m := \neg x_m]$)
- compute $g(\varphi')$

g is polynomial because if $f = O(n^k)$, then
 $g = O(n^{k+1})$